INCH-POUND
MIL-PRF-55182/11F
w/Amendment 1
15 August 2024
SUPERSEDING
MIL-PRF-55182/11F
13 October 2020

PERFORMANCE SPECIFICATION SHEET

RESISTOR, FIXED, FILM, ESTABLISHED RELIABILITY, STYLE RN*51 1/

This specification is Inactive for New Design after 19 November 1985.
For new design use MIL-PRF-55182/7.

This specification is approved for use by all Departments and agencies of the Department of Defense.

The requirements for acquiring the product described herein shall consist of this specification sheet and MIL-PRF-55182.

1. SCOPE

1.1 Scope. This specification covers the requirements for style RN*51 1/, established reliability, fixed, film resistors. Designers are CAUTIONED on using these resistors in high power pulse applications (see 6.6).

1.2 Part or Identifying Number (PIN). Resistors covered by this specification are identified by a PIN which is derived in accordance with MIL-PRF-55182 and is in the following form:

RN*51C1001FS 1/

RN*51 - Style and Termination type
C1001FS - Coded number

2. APPLICABLE DOCUMENTS

2.1 General. The documents listed in this section are specified in sections 3 and 4 of this specification. This section does not include documents cited in other sections of this specification or recommended for additional information or as examples. While every effort has been made to ensure the completeness of this list, document users are cautioned that they must meet all specified requirements documents cited in sections 3 and 4 of this specification, whether or not they are listed.

Comments, suggestions, or questions on this document should be addressed to: DLA Land and Maritime, ATTN: VAT, Post Office Box 3990, Columbus Ohio 43218-3990, or emailed to resistor@dla.mil. Since contact information can change, you may want to verify the currency of this address information using the ASSIST Online database at https://assist.dla.mil.

1/ Third letter is variable, dependent upon lead material or capability (see table I of MIL-PRF-55182).

AMSC N/A
FSC 5905

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2.2 Government documents.

2.2.1 Specifications, standards, and handbooks. The following specifications, standards, and handbooks form a part of this document to the extent specified herein. Unless otherwise specified, the issues of these documents are those cited in the solicitation or contract.

DEPARTMENT OF DEFENSE SPECIFICATION

MIL-PRF-55182	-	Resistor, Fixed, Film, Nonestablished Reliability, Established Reliability, and Space Level, General Specification for.

(Copies of these documents are available online at https://quicksearch.dla.mil)

2.3 Order of precedence. Unless otherwise noted herein or in the contract, in the event of a conflict between the text of this document and the references cited herein (except for related, specification sheets), the text of this document takes precedence. Nothing in this document, however, supersedes applicable laws and regulations unless a specific exemption has been obtained.

3. REQUIREMENTS

3.1 General. The requirements for acquiring the product described herein shall consist of this document and MIL-PRF-55182.

3.2 Interface and physical dimensions. The resistors shall meet the interface and physical dimensions specified in figure 1.

3.2.1 Characteristics. Style RN*51 is available in characteristics H, J, and K. 1/

3.2.2 Terminal types. Terminal type RNR is inactive for new design with characteristics H, J, and K. Terminal type RNN is inactive for new design when specified with characteristics H, J, and K (see 6.7).

3.3 Power rating. The power rating shall be 0.050 watt, based on full load operation at an ambient temperature of 125°C (see 6.4).

3.4 Voltage rating. The continuous working voltage shall not exceed 200 volts.

3.5 Resistance. The resistance values shall be 10 ohms minimum to 0.793 megohms maximum for characteristics H, J, and K.

3.6 Voltage coefficient (applicable to resistors of 1,000 ohms and above). The voltage coefficient shall not exceed ±0.005 percent per volt.

3.7 Mechanical sheer. Mechanical sheer is required for this style resistor.

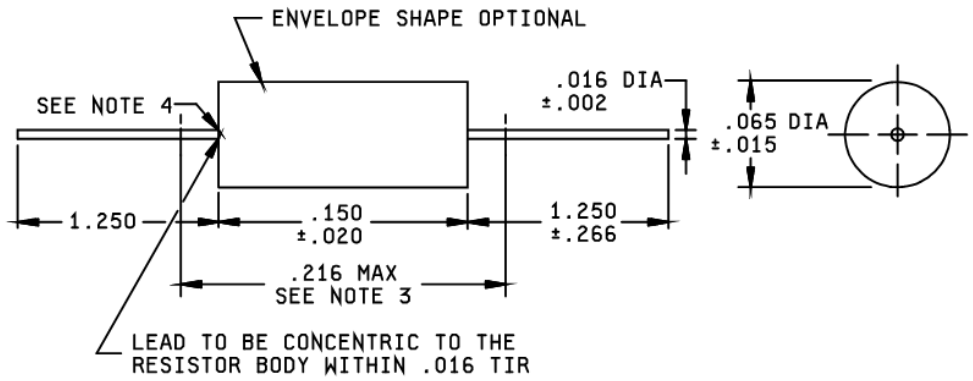
3.8 Insulation. Insulating material surrounding the resistor body shall extend around the entire shoulder of each end cap toward the axis of the lead attachment.

3.9 Marking. Due to size limitations this style resistor shall be marked with the following minimum information:

226K	-	Date code, characteristic
10R0	-	Coded resistance
FMCJ	-	Tolerance, failure rate, terminal type and JAN marking

1/ Third letter is variable, dependent upon lead material or capability (see table I of MIL-PRF-55182).

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Inches	mm	Inches	mm	Inches	mm
0.002	0.05	0.020	0.51	0.216	5.49
0.015	0.38	0.065	1.65	0.266	6.76
0.016	0.41	0.150	3.81	1.250	31.75

NOTES:
1. Dimensions are in inches.
2. Metric equivalents are given for general information.
3. The maximum length is “clean lead” to “clean lead”.
4. The end of the body shall be that point at which the body diameter equals the nearest drill size larger than 250 percent of the nominal lead diameter.
5. Lead length for new design and tape real packaging shall be 1.00 inch +0.625 inch, -0.000 inch (25.4 mm +15.80 mm, -0.00 mm).
6. Lead concentric tolerance is to be measured at the point of lead egress from the resistor body.

FIGURE 1. Style RN*51 resistor. 1/

3.10 Pure tin. The use of pure tin, as an underplate or final finish is prohibited both internally and externally. Tin content of resistor components and solder shall not exceed 97 percent, by mass. Tin shall be alloyed with a minimum of 3 percent lead, by mass (see 6.9).

4. VERIFICATION

4.1 Verification. Verification shall be in accordance with MIL-PRF-55182.

5. PACKAGING

5.1 Packaging. For acquisition purposes, the packaging requirements shall be as specified in the contract or order (see 6.2). When packaging of material is to be performed by DoD or in-house contractor personnel, these personnel need to contact the responsible packaging activity to ascertain packaging requirements. Packaging requirements are maintained by the Inventory Control Point's packaging activities within the Military Service or Defense Agency, or within the military service’s system commands. Packaging data retrieval is available from the managing Military Department's or Defense Agency's automated packaging files, CD-ROM products, or by contacting the responsible packaging activity.

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6. NOTES

(This section contains information of a general or explanatory nature that may be helpful, but is not mandatory.)

6.1 Intended use. The intended use specified in MIL-PRF-55182 is applicable to this specification.

6.2 Acquisition requirements. Acquisition documents must specify the following:

a. Title, number, and date of this specification, and the complete PIN (see 1.2).
b. Unless otherwise specified (see 2.1), the versions of the individual documents referenced will be those in effect on the date of release of the solicitation.
c. Packaging requirements (see 5.1). (i.e. Electrostatic discharge (ESD) sensitive packaging).

6.3 Reliability levels. Data submitted for reliability levels established for MIL-PRF-55182/7 may be combined with this specification data to establish or maintain the reliability level of this specification if identical resistors are both qualified for MIL-PRF-55182/7 and this specification.

6.4 Power rating at 70°C. The power at 70°C for 2,000 hours life test has been established at 0.1 watt, maximum voltage is 200 volts. It should be noted that the failure rate level is established at the single condition of 125°C, 0.05 watt and 10,000 hours life test duration.

6.5 Pulse applications. Designers are CAUTIONED on using these resistors in high power pulse applications. Since they have not been qualified nor tested for such applications, damage and premature failure are possible. This resistor only sees a one hour overload (see 4.8.3 of MIL-PRF-55182) as part of the group A inspection of this specification. Designers MAY CONSIDER using DLA Land and Maritime drawing 03003 for high power pulse applications. NOTE: This resistor does not have the geometry (form, fit) of the MIL-PRF-55182 resistor, nor are they subject to the same Qualification/verification, Conformance inspection, or Periodic Group C inspection requirements.

6.6 RNN terminal substitution. Due to the loss of the last source of supply for weldable RNN style resistors with characteristic H, J, and K, it is recommended instead to use the RNC style resistors with solderable/weldable terminals for equivalent characteristic, resistance value, tolerance, and product level.

6.7 Failure Rate (FR) level supersession (see 3.32.4.1 of MIL-PRF-55182). **FOR DLA ACQUISITION PURPOSES ONLY**. In the event the manufacturer has discontinued a FR level (e.g. “S”) and is qualified to a higher FR level (e.g. “R”), it is recommended that the next available FR level be used for acquisition. This can only be used if the FR is no longer available by the manufacturer(s) listed in the Qualified Product List (QPL) for MIL-PRF-55182. On multiple QPL sourced items, this will not apply if a manufacturer is still qualified to the lower FR level.

6.8 Electrostatic charge. Under several combinations of conditions, these resistors can be electrically damaged, by electrostatic charges, and drift from specified value. Users should consider this phenomena when ordering or shipping resistors. Direct shipment to the Government is controlled by MIL-DTL-39032 which specifies a preventive packaging procedure.

6.9 Tin whisker growth. The use of alloys with tin content greater than 97 percent, by mass, may exhibit tin whisker growth problems after manufacture. Tin whiskers may occur anytime from a day to years after manufacture and can develop under typical operating conditions, on products that use such materials. Conformal coatings applied over top of a whisker-prone surface will not prevent the formation of tin whiskers. Alloys of tin containing 3 percent or more lead, by mass, have shown to inhibit the growth of tin whiskers. For additional information on this matter, refer to ASTM-B545 (Standard Specification for Electrodeposited Coatings of Tin).

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6.10 Amendment notations. The margins of this specification are marked with vertical lines to indicate modification generated by this amendment. This was done as a convenience only and the Government assumes no liability whatsoever for any inaccuracies in these notations. Bidders and contractors are cautioned to evaluate the requirements of this document based on the entire content irrespective of the marginal notations and relationship.

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(Project 5905-2024-052)

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